



CifloxinTM

INFUSION (2 MG/ML)
SOLUTION FOR INFUSION

PRODUCT NAME
CIFLOXIN INFUSION (2 MG/ML) (SOLUTION FOR INFUSION)

NAME AND STRENGTH OF ACTIVE SUBSTANCE
Each glass bottle with 100 mL infusion solution contains 200 mg of ciprofloxacin.
Each glass bottle with 200 mL infusion solution contains 400 mg of ciprofloxacin.

PRODUCT DESCRIPTION
Clear, colorless sterile solution (pH 3.5 - 4.6)

PHARMACODYNAMICS
Pharmacotheapeutic group: fluoroquinolones
Mechanism of Action
As a fluoroquinolone antibacterial agent, the bactericidal action of ciprofloxacin results from the inhibition of both type II topoisomerase (DNA gyrase) and topoisomerase IV, required for bacterial DNA replication, transcription, repair and recombination.
Mechanism of Resistance
In vitro resistance to ciprofloxacin can be acquired through a stepwise process by target site mutations in both DNA gyrase and topoisomerase IV. The degree of cross-resistance between ciprofloxacin and other fluoroquinolones that results is variable. Single mutations may not result in clinical resistance, but multiple mutations generally result in clinical resistance to many or all active substances within the class.
Impermeability and/or active substance efflux pump mechanisms of resistance may have a variable effect on susceptibility to fluoroquinolones, which depends on the physicochemical properties of the various active substances within the class and the affinity of transport systems for each active substance.
All in-vitro mechanisms of resistance are commonly observed in clinical isolates. Resistance mechanisms that inactivate other antibiotics such as permeation barriers (common in *Pseudomonas aeruginosa*) and efflux mechanisms may affect susceptibility to ciprofloxacin. Plasmid-mediated resistance encoded by qnr-genes has been reported.
Microbiological Susceptibility
The prevalence of acquired resistance may vary geographically and time dependent for selected species, local information on resistance is desirable, particularly when treating severe infections. Ciprofloxacin is usually active against the following microorganisms *in vitro*:

COMMONLY SUSCEPTIBLE SPECIES		
Gram-Positive Aerobes		
<i>Bacillus anthracis</i> (1)		
Gram-Negative Aerobes		
<i>Aeromonas</i> spp.	<i>Haemophilus influenzae</i> *	<i>Salmonella</i> spp.*
<i>Bacillus</i> spp.	<i>Legionella</i> spp.	<i>Shigella</i> spp.*
<i>Citrobacter koseri</i>	<i>Moraxella catarrhalis</i> *	<i>Vibrio</i> spp.
<i>Francisella tularensis</i>	<i>Neisseria meningitidis</i>	<i>Yersinia pestis</i>
<i>Haemophilus ducreyi</i>	<i>Pasteurella</i> spp.	
Anaerobes		
<i>Mobiluncus</i>		
Others		
<i>Chlamydia trachomatis</i> (5)	<i>Mycoplasma hominis</i> (5)	
<i>Chlamydia pneumoniae</i> (5)	<i>Mycoplasma pneumoniae</i> (5)	
ORGANISMS FOR WHICH ACQUIRED RESISTANCE MAY BE A PROBLEM		
Gram-Positive Aerobes		
<i>Enterococcus faecalis</i> (5)	<i>Staphylococcus</i> spp. (12)	
Gram-Negative Aerobes		
<i>Acinetobacter baumannii</i> *	<i>Escherichia coli</i> *	<i>Proteus vulgaris</i> *
<i>Burkholderia cepacia</i> **	<i>Klebsiella oxytoca</i>	<i>Providencia</i> spp.
<i>Campylobacter</i> spp.**	<i>Klebsiella pneumoniae</i> *	<i>Pseudomonas aeruginosa</i> *
<i>Citrobacter freundii</i> *	<i>Morganella morganii</i> *	<i>Pseudomonas fluorescens</i>
<i>Enterobacter aerogenes</i>	<i>Neisseria gonorrhoeae</i> *	<i>Serratia marcescens</i> *
<i>Enterobacter cloacae</i> *	<i>Proteus mirabilis</i> *	
Anaerobes		
<i>Peptostreptococcus</i> spp.	<i>Propionibacterium acnes</i>	
INHERENTLY RESISTANT ORGANISMS		
Gram-Positive Aerobes		
<i>Actinomyces</i>	<i>Listeria monocytogenes</i>	<i>Enterococcus faecium</i>
Gram-Negative Aerobes		
<i>Stenotrophomonas maltophilia</i>		
Anaerobes		
Excepted as listed above		
Others		
<i>Mycoplasma genitalium</i>	<i>Ureaplasma urealyticum</i>	
* Clinical efficacy has been demonstrated for susceptible isolates in approved clinical indications		
* Resistance rate $\geq$ 50% in one or more EU countries		
(5): Natural intermediate susceptibility in the absence of acquired mechanism of resistance		
(1): Studies have been conducted in experimental animal infections due to inhalations of <i>Bacillus anthracis</i> spores; these studies reveal that antibiotics starting early after exposition avoid the occurrence of the disease if the treatment is made up to the decrease of the number of spores in the organism under the infective dose. The recommended use in human subjects is based primarily on in-vitro susceptibility and on animal experimental data together with limited human data. Two-month treatment duration in adults with oral ciprofloxacin given at the following dose, 500 mg bid, is considered as effective to prevent anthrax infection in humans. The treating physician should refer to national and /or international consensus documents regarding treatment of anthrax.		
(2): Methicillin-resistant <i>S. aureus</i> very commonly express co-resistance to fluoroquinolones. The rate of resistance to methicillin is around 20 to 50% among all staphylococcal species and is usually higher in nosocomial isolates.		

PHARMACOKINETICS
Absorption
Following an intravenous (IV) infusion of ciprofloxacin the mean maximum serum concentration (C_{max}) were achieved at the end of infusion. Pharmacokinetics of ciprofloxacin was linear over the dose range up to 400 mg administered intravenously.
Comparison of the pharmacokinetic parameters for a twice a day and three times a day IV dose regimen indicated no evidence of drug accumulation for ciprofloxacin and its metabolites.
A 60 minutes IV infusion of 200 mg ciprofloxacin or the oral administration of 250 mg ciprofloxacin, both given every 12 hours, produced an equivalent area under the serum concentration time curve (AUC).
A 60 minutes IV infusion of 400 mg ciprofloxacin every 12 hours was bioequivalent to 500 mg oral dose every 12 hours with regards to AUC.
The 400 mg IV dose administered over 60 minutes every 12 hours resulted in a C_{min} similar to that observed with a 750 mg oral dose. A 60 minutes infusion of 400 mg ciprofloxacin every 8 hours is equivalent with respect to AUC to 750 mg oral regimen given every 12 hours.
Distribution
Protein binding of ciprofloxacin is low (20 - 30%). Ciprofloxacin is present in plasma largely in a non-ionized form and has a large steady state distribution volume of 2 - 3 L/kg body weight. Ciprofloxacin reaches high concentrations in a variety of tissues such as lung (epithelial fluid, alveolar macrophages, biopsy tissue), sinuses, inflamed lesions (cantharides blister fluid), and the urogenital tract (urine, prostate, endometrium) where total concentrations exceeding those of plasma concentrations are reached.
Biotransformation
Low concentrations of four metabolites have been reported, which were identified as: desethylenciprofloxacin (M₁), sulphociprofloxacin (M₂), oxociprofloxacin (M₃) and formylciprofloxacin (M₄). The metabolites display in vitro antimicrobial activity but to a lower degree than the parent compound.
Ciprofloxacin is known to be a moderate inhibitor of the CYP 450 1A2 iso-enzymes.
Elimination
Ciprofloxacin is largely excreted unchanged both renally and, to a smaller extent, fecally. Renal clearance is between 180 - 300 mL/kg/h and the total body clearance is between 480 - 600 mL/kg/h. Ciprofloxacin undergoes both glomerular filtration and tubular secretion. Severely impaired renal function leads to increased half-lives of ciprofloxacin of up to 12 hours.
Non-renal clearance of ciprofloxacin is mainly due to active trans-intestinal secretion and metabolism. 1% of the dose is excreted via the biliary route. Ciprofloxacin is present in the bile in high concentrations.
Paediatric patients
The pharmacokinetic data in paediatric patients are limited. Based on population pharmacokinetic analysis of paediatric patients with various infections, the predicted mean half-life in children is approx. 4 - 5 hours and the bioavailability of the oral suspension ranges from 50 to 80%.

INDICATION
Consideration should be given to applicable official guidance on the appropriate use of antibacterial agents.
Uncomplicated and complicated infections caused by ciprofloxacin sensitive pathogens:

- Infections of the respiratory tract*
- Ciprofloxacin can be regarded as an advisable treatment for pneumonias caused by *Klebsiella*, *Enterobacter*, *Proteus*, *E. coli*, *Pseudomonas*, *Haemophilus*, *Moraxella catarrhalis*, *Legionella* and *Staphylococcus*
- Infections of the middle ear (otitis media), of the paranasal sinuses (sinusitis)* especially if these are caused by Gram-negative organisms including *Pseudomonas aeruginosa* or by staphylococci
- Infections of the eyes
- Infections of the kidneys and/or the efferent urinary tract*
- Infections of the genital organs, including adnexitis, gonorrhoea, prostatitis*
- Infections of the abdominal cavity (e.g. infections of the gastrointestinal tract or of the biliary tract, peritonitis)*
- Infections of the skin and soft tissue*
- Infections of the bones and joints*
- Sepsis*
- Infections or imminent risk of infection (prophylaxis) in patients whose immune system has been weakened (e.g. patients on immunosuppressants or have neutropenia)
- Selective intestinal decontamination in immunosuppressed patients

*CIFLOXIN INFUSION (2 MG/ML) (SOLUTION FOR INFUSION) should only be used:
• When *Pseudomonas* is considered AND the patient is allergic to antipseudomonal penicillins/cephalosporins;
• For resistant organisms with no other alternative antibiotics available

RECOMMENDED DOSAGE	
General Dosage Recommendations	
Adults	
Indication	Recommended Dosage (IV infusion)
Respiratory Tract Infection (according to severity and organism)	2 x 400 mg to 3 x 400 mg
Urinary Tract Infection • acute, uncomplicated • complicated	2 x 200 mg to 2 x 400 mg 2 x 400 mg to 3 x 400 mg
Genital Infection • adnexitis, prostatitis, epididymo-orchitis	2 x 400 mg to 3 x 400 mg
Other Infections (See Indication)	2 x 400 mg
Particularly severe, life threatening infections • recurrent infections in cystic fibrosis • bone and joint infections • septicemia • peritonitis	3 x 400 mg
Special Populations	
Children and Adolescents	
Cystic Fibrosis	
• 10 mg/kg body weight intravenous (IV) three times daily with a maximum of 400 mg per dose.	
• Complicated Urinary Tract Infections and Pyelonephritis	
• 6 - 10 mg/kg body weight IV three times daily with a maximum of 400 mg per dose	
Geriatric Patients (> 65 years)	
• Elderly patients should receive a dose as low as possible depending on the severity of their illness and the creatinine clearance	

Renal and Hepatic Impairment			
Renal/Liver Function	Creatinine Clearance [mL/min/1.73m ²]	Serum Creatinine [mg/100mL]	Recommended Dose (IV Infusion)
Impaired Renal Function	30 - 60	1.4 - 1.9	Maximum 800 mg daily
	< 30	$\geq$ 2.0	Maximum 400 mg daily
Impaired Renal Function on Haemodialysis	30 - 60	1.4 - 1.9	Maximum 800 mg daily
	< 30	$\geq$ 2.0	Maximum 400 mg daily (on dialysis days after dialysis)
Impaired renal function on continuous ambulatory peritoneal dialysis (CAPD)	-	-	Addition of ciprofloxacin infusion solution to the dialysate (intrapertoneal): 50 mg ciprofloxacin / liter dialysate given every 6 hours
Impaired liver function	-	-	No dose adjustment is required
Impaired renal and liver function	30 - 60	1.4 - 1.9	Maximum 800 mg daily
	< 30	$\geq$ 2.0	Maximum 400 mg daily

[Note: Dosing in children with impaired renal and or hepatic function has not been studied]

ROUTE OF ADMINISTRATION
Intravenous infusion
Method of administration
If the patient is unable to take ciprofloxacin film-coated tablets because of the severity of the illness or for other reasons (e.g. patients on enteral nutrition), it is recommended to commence the therapy with an intravenous form of ciprofloxacin. After intravenous administration, the treatment can be continued orally.
Intravenous infusion
Ciprofloxacin should be administered by intravenous infusion over a period of 60 minutes. Slow infusion into a large vein will minimize patient discomfort and reduce the risk of venous irritation. The infusion solution can be infused either directly or after mixing with other compatible infusion solutions (See *Incompatibilities*).
Duration of treatment
The duration of treatment depends on the severity of the illness and on the clinical and bacteriological course. It is essential to continue therapy for at least 3 days after disappearance of the fever or of the clinical symptoms.
Mean duration of treatment

- 1 day for acute uncomplicated gonorrhea and cystitis
- Up to 7 days for infections of the kidneys, urinary tract and abdominal cavity
- Over the entire period of the neutropenic phase in patients with weakened body defenses
- A maximum of 2 months in osteomyelitis
- 7 - 14 days in all other infections
- At least 10 days for streptococcal infections due to the risk of late complications
- A minimum of 10 days for infections caused by *Chlamydia* spp.

Children and adolescents
Cystic fibrosis, for broncho-pulmonary infections associated with *Pseudomonas aeruginosa* in pediatric patients (aged 5 - 17 years), the duration of treatment is 10 - 14 days
Complicated urinary tract infections and pyelonephritis due to *Escherichia coli*, the duration of treatment is 10 - 21 days

CONTRAINDICATIONS
Hypersensitivity to the active substance, to other quinolones or to any of the excipients.
Concomitant administration of ciprofloxacin and tizanidine (see *Interaction with Other Medicinal Products / Other Forms of Interaction*).

WARNINGS AND PRECAUTIONS
The use of ciprofloxacin should be avoided in patients who have experienced serious adverse reactions in the past when using fluoroquinolones containing products (see *Undesirable Effects*). Treatment of these patients with ciprofloxacin should only be initiated in the absence of alternative treatment options and after careful benefit / risk assessment.
Aortic aneurysm and dissection
Epidemiologic studies report an increased risk of aortic aneurysm and dissection after intake of fluoroquinolones, particularly in the older population.
Therefore, fluoroquinolones should only be used after careful benefit-risk assessment and after consideration of other therapeutic options in patients with positive family history of aneurysm disease, or in patients diagnosed with pre-existing aortic aneurysm and/or aortic dissection, or in presence of other risk factors or conditions predisposing for aortic aneurysm and dissection (e.g. Marfan syndrome, vascular Ehlers-Danlos syndrome, Takayasu arteritis, giant cell arteritis, Behcet's disease, hypertension, known atherosclerosis). In case of sudden abdominal, chest or back pain, patients should be advised to immediately consult a physician in an emergency department.
Psychiatric reactions
Psychiatric reactions may occur even after the first administration of fluoroquinolones, including Cifloxin Infusion. In rare cases, depression or psychotic reactions can progress to suicidal ideation/thoughts and self-injurious behaviour, such as attempted or completed suicide (see section 'Undesirable effects'). In the event that the patient develops such reactions, Cifloxin Infusion should be discontinued and appropriate measures instituted. Caution is recommended if Cifloxin Infusion is to be used in psychotic patients or in patients with a history of psychiatric disease.
Prolonged, disabling and potentially irreversible serious adverse drug reactions
Very rare cases of prolonged (continuing months or years), disabling and potentially irreversible serious adverse drug reactions affecting different, sometimes multiple body systems (musculoskeletal, nervous, psychiatric and senses) have been reported in patients receiving fluoroquinolones irrespective of their age and pre-existing risk factors. Ciprofloxacin should be discontinued immediately at the first signs or symptoms of any serious adverse reaction and patients should be advised to contact their prescriber for advice.
Tendinitis and tendon rupture
Tendinitis and tendon rupture (especially but not limited to Achilles tendon), sometimes bilateral, may occur as early as within 48 hours of starting treatment with fluoroquinolones and have been reported to occur even up to several months after discontinuation of treatment. The risk of tendinitis and tendon rupture is increased in older patients (above 60 years of age), with renal impairment, patients with solid organ transplants, and those treated concurrently with corticosteroids. Therefore, concomitant use of corticosteroids should be avoided. At the first sign of tendinitis (e.g. painful swelling, inflammation) the treatment with ciprofloxacin should be discontinued and alternative treatment should be considered. The affected limb(s) should be appropriately treated (e.g. immobilisation). Corticosteroids should not be used if signs of tendinopathy occur.
Peripheral neuropathy
Cases of sensory or sensorimotor polyneuropathy resulting in paraesthesia, hypaesthesia, dysesthesia, or weakness have been reported in patients receiving quinolones and fluoroquinolones. Patients under treatment with ciprofloxacin should be advised to inform their doctor and pharmacist prior to continuing treatment if symptoms of neuropathy such as pain, burning, tingling, numbness, or weakness develop in order to prevent the development of potentially irreversible condition (see *Undesirable Effects*).
Exacerbation of myasthenia gravis
Fluoroquinolones have neuromuscular blocking activity and may exacerbate muscle weakness in person with myasthenia gravis. Post marketing serious adverse events, including deaths and requirement for ventilator support have been associated with fluoroquinolones use in persons with myasthenia gravis. Avoid fluoroquinolones in patients with known history of myasthenia gravis.

Actual Size 100 %

Drawing No. CIF-INF-100-200-03-L-ENG-MY-A0003

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back

29.7 cm

21 cm

Severe infections and mixed infections with gram-positive and anaerobic pathogens

Ciprofloxacin monotherapy are not suited for treatment of severe infections and infections that might be due to Gram-positive or anaerobic pathogens. In such infections, ciprofloxacin must be co-administered with other appropriate antibacterial agents.

Streptococcal infections (including *Streptococcus pneumoniae*)

Ciprofloxacin is not recommended for the treatment of streptococcal infections due to inadequate efficacy.

Genital tract infections

Epididymo-orchitis and pelvic inflammatory diseases may be caused by fluoroquinolone-resistant *Neisseria gonorrhoeae* isolates. For epididymo-orchitis and pelvic inflammatory diseases, empirical ciprofloxacin should only be considered in combination with another appropriate antibacterial agent (e.g. a cephalosporin) unless ciprofloxacin-resistant *Neisseria gonorrhoeae* can be excluded. If clinical improvement is not achieved after 3 days of treatment, the therapy should be reconsidered.

NaCl load

In patients for whom sodium intake is of medical concern (e.g. patients with congestive heart failure, renal failure, nephrotic syndrome, etc.), the additional sodium load should be taken into account.

Hypersensitivity

Hypersensitivity and allergic reactions, including anaphylaxis and anaphylactoid reactions, may occur following a single dose and may be life threatening. If such reaction occurs, ciprofloxacin should be discontinued and an adequate medical treatment is required.

Photosensitivity

Ciprofloxacin has been shown to cause photosensitivity reactions. Patients taking ciprofloxacin should be advised to avoid direct exposure to either extensive sunlight or UV irradiation during treatment.

Gastrointestinal system

The occurrence of severe and persistent diarrhoea during or after treatment (including several weeks after treatment) may indicate an antibiotic-associated colitis (life threatening with possible fatal outcome), requiring immediate treatment. In such cases, ciprofloxacin should immediately be discontinued, and an appropriate therapy initiated. Anti-peristaltic drugs are contraindicated in this situation.

Injection site reaction

Local intravenous (IV) site reactions have been reported following IV administration of ciprofloxacin. These reactions are more frequent if the infusion time is 30 minutes or less. These may appear as local skin reactions which resolve rapidly upon completion of the infusion. Subsequent IV administration is not contraindicated unless the reactions recur or worsen.

Paediatric population

The use of ciprofloxacin in children and adolescents should follow available official guidance. Ciprofloxacin treatment should be initiated only by physicians who are experienced in the treatment of cystic fibrosis and/or severe infections in children and adolescents. Treatment should be initiated only after a careful benefit/risk evaluation, due to possible adverse events related to joints and/or surrounding tissue.

Interaction with tests

Ciprofloxacin in vitro potency may interfere with the *Mycobacterium* spp. culture test by suppression of mycobacterial growth, causing false negative results in specimens from patients currently taking ciprofloxacin.

INTERACTION WITH OTHER MEDICINAL PRODUCTS / OTHER FORMS OF INTERACTION

Theophylline

Concurrent administration of ciprofloxacin and theophylline can cause an undesirable increase in serum theophylline concentration. This can lead to theophylline-induced side effects that may rarely be life threatening or fatal. During the combination, serum theophylline concentrations should be checked and the theophylline dose reduced as necessary.

Methotrexate

Renal tubular transport of methotrexate may be inhibited by concomitant administration of ciprofloxacin, potentially leading to increased plasma levels of methotrexate and increased risk of methotrexate-associated toxic reactions. The concomitant use is not recommended.

Cyclosporin

A transient rise in the concentration of serum creatinine was observed when ciprofloxacin and cyclosporin were administered simultaneously. Therefore, it is necessary to control the patient serum creatinine concentrations frequently (twice a week).

Probenecid

Probenecid interferes with renal secretion of ciprofloxacin. Co-administration of probenecid and ciprofloxacin increases the ciprofloxacin serum concentrations.

Tizanidine

Serum tizanidine concentration increased when given concomitantly with ciprofloxacin. The increased of tizanidine plasma concentration is associated with a potentiated hypotensive and sedative effect. Therefore, tizanidine must not be administered together with ciprofloxacin.

Duloxetine

Concomitant use of duloxetine with strong inhibitors of the CYP450 1A2 isozyme such as fluvoxamine, may result in an increase serum concentration of duloxetine. Although no clinical data are available on a possible interaction with ciprofloxacin, similar effects can be expected upon concomitant administration.

Clozapine

Serum concentrations of clozapine and N-desmethylclozapine were increased, following concomitant administration of 250 mg ciprofloxacin with clozapine for 7 days. Clinical surveillance and appropriate adjustment of clozapine dosage during and shortly after co-administration with ciprofloxacin are advised.

Other xanthine derivatives

On concurrent administration of ciprofloxacin and caffeine or pentoxifylline (xopentifylline) containing products, raised serum concentrations of these xanthine derivatives were reported.

Vitamin K antagonists

Simultaneous administration of ciprofloxacin with a vitamin K antagonist may augment its anticoagulant effects. The risk may vary with the underlying infection, age and general status of the patient so that the contribution of ciprofloxacin to the increase in INR (international normalized ratio) is difficult to assess. The INR should be monitored frequently during and shortly after co-administration of ciprofloxacin with a vitamin K antagonist (e.g. warfarin, acenocoumarol, phenprocoumon, or flutidione).

Sildenafil

Serum concentration of sildenafil was increased approximately 2-fold in healthy subjects after an oral dose of 50 mg given concomitantly with 500 mg ciprofloxacin. Therefore, caution should be used when prescribing ciprofloxacin concomitantly with sildenafil, taking into consideration the risks and the benefits.

PREGNANCY AND LACTATION

Pregnancy

It cannot be excluded that the drug could cause damage to articular cartilage in the human immature organism / foetus. As a precautionary measure, ciprofloxacin should not be used during pregnancy.

Lactation

Ciprofloxacin is excreted in breast milk. Due to the potential risk of articular damage, ciprofloxacin should not be used during breast-feeding.

EFFECTS ON ABILITY TO DRIVE AND USE MACHINE

Due to its neurological effects, ciprofloxacin may affect reaction time. Thus, the ability to drive or to operate machinery may be impaired.

UNDESIRABLE EFFECTS

Summary of Safety Profile

The most common reported adverse reactions are nausea, diarrhoea, vomiting, transient increase in transaminases, rash, injection and infusion site reactions. The adverse reactions presented below are classified by system organ from both oral and intravenous administration of ciprofloxacin.

System Organ	Frequency	Adverse Reactions
Infections and infestations	Uncommon	: Mycotic superinfections
Blood and lymphatic system disorders	Uncommon	: Eosinophilia
	Rare	: Leukopenia, neutropenia, thrombocytopenia, thrombocytaemia, leukocytosis, anaemia
	Very rare	: Haemolytic, anaemia, agranulocytosis, bone marrow depression and pancytopenia (<i>life-threatening</i>)
	Rare	: Allergic reaction, allergic oedema/angioedema
Immune system disorders	Very rare	: Anaphylactic reaction, serum sickness-like reaction, anaphylactic shock (<i>life-threatening</i>)
	Uncommon	: Decreased appetite
Metabolism and nutrition disorders	Rare	: Hyperglycaemia, hypoglycaemia
Psychiatric disorders*	Uncommon	: Psychomotor hyperactivity/agitation
	Rare	: Confusion and disorientation, anxiety reaction, abnormal dreams, depression (potentially culminating in suicidal ideation / thoughts or suicide attempts and completed suicide), hallucination
	Very rare	: Psychotic reactions (potentially culminating in suicidal ideations / thoughts or suicide attempts and completed suicide)
	Not known	: Mania, including hypomania
Nervous system disorders*	Uncommon	: Headache, taste disorders, sleep disorders, dizziness
	Rare	: Par- and Dysaesthesia, hypoaesthesia, tremor, vertigo, seizures (including status epilepticus)
	Very rare	: Migraine, disturbed coordination, smell disorders, intracranial hypertension and pseudotumor cerebri
	Not known	: Peripheral neuropathy and polyneuropathy
Eye disorders*	Rare	: Visual disturbances (e.g. diplopia)
	Very rare	: Visual colour distortions
Ear and labyrinth disorders*	Rare	: Tinnitus, hearing loss/impaired
Cardiac disorders	Rare	: Tachycardia
	Not known	: Ventricular arrhythmia, ECG QT prolonged, torsades de pointes (<i>predominant in patient with risk for QT prolongation</i>)
Vascular disorders	Rare	: Vasodilatation, hypotension, syncope
	Very rare	: Vasculitis
Respiratory, thoracic and mediastinal disorders	Rare	: Dyspnoea (including asthmatic condition)
Gastrointestinal disorders	Common	: Nausea, diarrhoea
	Uncommon	: Vomiting, gastrointestinal and abdominal pains, dyspepsia, flatulence
	Rare	: Antibiotic associated colitis
Hepatobiliary disorders	Very rare	: Pancreatitis
	Uncommon	: Increase in transaminases, increased bilirubin
	Rare	: Hepatic impairment, jaundice, hepatitis (non-infective)
Skin and subcutaneous tissue disorders	Very rare	: Liver necrosis (very rarely progressing to life threatening hepatic failure)
	Uncommon	: Rash, pruritus, urticarial
	Rare	: Photosensitivity reactions
	Very rare	: Petechiae, erythema multiforme, erythema nodosum, Stevens-Johnson syndrome, toxic epidermal necrolysis
	Not known	: Acute Generalized Exanthematous Pustulosis (AGEP), Drug Reaction with Eosinophilia and Systemic Symptoms (DRESS)
Musculoskeletal and connective tissue disorders*	Uncommon	: Arthralgia, musculoskeletal pain
	Rare	: Myalgia, arthritis, increased muscle tone and cramp
	Very rare	: Muscular weakness, tendinitis, tendon rupture, exacerbation of myasthenia gravis (<i>post marketing experience</i>)
Renal and urinary disorders	Uncommon	: Renal Impairment
	Rare	: Renal failure, tubulointerstitial nephritis, haematuria, crystalluria
General disorders and administration site conditions*	Common	: Injection and infusion site reactions (only IV route)
	Uncommon	: Asthenia, fever
	Rare	: Oedema, sweating (hyperhidrosis)
Investigations	Uncommon	: Increase in blood alkaline phosphatase
	Rare	: Increased amylase
	Not known	: International normalized ratio increased (in patients treated with Vitamin K antagonists)

Remark

*Very rare cases of prolonged (up to months or years), disabling and potentially irreversible serious drug reactions affecting several, sometimes multiple, system organ classes and senses (including reactions such as tendinitis, tendon rupture, arthralgia, pain in extremities, gait disturbance, neuropathies associated with paraesthesia, depression, fatigue, memory impairment, sleep disorders, and impairment of hearing, vision, taste and smell) have been reported in association with the use of fluoroquinolones in some cases irrespective of pre-existing risk factors (see *Warnings and Precautions*).

The following undesirable effects have a higher frequency category in the subgroups of patients receiving intravenous or sequential (intravenous to oral) treatment:

Common	Vomiting, transient increase in transaminases, rash
Uncommon	Thrombocytopenia, thrombocytaemia, confusion and disorientation, hallucinations, par- and dysaesthesia, seizures, vertigo, visual disturbances, hearing loss, tachycardia, vasodilatation, hypotension, transient hepatic impairment, jaundice, renal failure, oedema
Rare	Pancytopenia, bone marrow depression, anaphylactic shock, psychotic reactions, migraine, smell disorders, hearing impaired, vasculitis, pancreatitis, liver necrosis, petechiae, tendon rupture

Paediatric Population

The incidence of arthropathy, mentioned above, is referring to data collected in studies with adults. In children, arthropathy is reported to occur commonly.

OVERDOSE AND TREATMENT

An overdose of 12 g has been reported to lead to mild symptoms of toxicity. An acute overdose of 16 g has been reported to cause acute renal failure.

Symptoms in overdose consist of dizziness, tremor, headache, tiredness, seizures, hallucinations, confusion, abdominal discomfort, renal and hepatic impairment as well as crystalluria and haematuria. Reversible renal toxicity has been reported.

Apart from routine emergency measures (e.g. ventricular emptying followed by medical carbon), it is recommended to monitor renal function, including urinary pH and acidity, if required, to prevent crystalluria. Patients should be kept well hydrated. Calcium or magnesium containing antacids may theoretically reduce the absorption of ciprofloxacin in overdoses.

Only a small quantity of ciprofloxacin (<10%) is eliminated by haemodialysis or peritoneal dialysis. In the event of overdose, symptomatic treatment should be implemented. ECG monitoring should be undertaken, because of the possibility of QT interval prolongation.

COMPATIBILITIES

The ciprofloxacin infusion solution is compatible with Ringer solution, Ringer lactate solution, 5% and 10% glucose solutions, 5% and 10% fructose solutions. When ciprofloxacin infusion solutions are mixed with compatible infusion solutions, for microbial reasons and light sensitivity these solutions must be administered shortly after admixture.

INCOMPATIBILITIES

This medicinal product must not be mixed with other medicinal products, unless compatibility with other infusion solutions/drugs has been confirmed, the infusion solution must always be administered separately. Examples of the visual signs of incompatibility are precipitation, clouding, and discoloration.

Incompatibility appears with all infusion solutions/drugs that are physically or chemically unstable at the pH of the solutions (e.g. penicillins, heparin solutions), especially in combination with solutions adjusted to an alkaline pH (pH-value of Cifloxin Infusion ranges from 3.5 to 4.6).

STORAGE CONDITIONS

Do not store above 30°C.

Protect from light and avoid freezing.

Keep the vial in the packaging box until time of use in order to protect from light.

SHELF LIFE

Refer to unit box for shelf-life.

DOSAGE FORMS AND PACKAGING AVAILABLE

Cifloxin Infusion (200 mg per 100 mL) : 1 glass bottle of 100 mL sterile solution per unit box

Cifloxin Infusion (400 mg per 200 mL) : 1 glass bottle of 200 mL sterile solution per unit box

DATE OF REVISION OF PI

Date of revision: 10/10/2025

Manufacturer:- SIAM BHEASACH CO., LTD.

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